

Master MVA

Optimization Reminders

Part II

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Iterating projections

Feasibility problem

Problem

Let $\mathcal{H}$ be a (finite) Hilbert space. Let $m \in \mathbb{N} \setminus \{0, 1\}$.

Let $(C_i)_{1 \leq i \leq m}$ be closed convex subsets of $\mathcal{H}$ such that $\bigcap_{i=1}^m C_i \neq \emptyset$.

We want to

$$\text{Find } \hat{x} \in \bigcap_{i=1}^m C_i.$$

POCS (Projection Onto Convex Sets) algorithm

For every $n \in \mathbb{N}$, let $i_n - 1$ denote the remainder after division of n by m .

Set $x_0 \in \mathcal{H}$

For $n = 0, \dots$

$$\quad \lfloor x_{n+1} = P_{C_{i_n}}(x_n).$$

Feasibility problem

(over-relaxation $\neq$ under-relaxation)

Problem

Let $\mathcal{H}$ be a (finite) Hilbert space. Let $m \in \mathbb{N} \setminus \{0, 1\}$.

Let $(C_i)_{1 \leq i \leq m}$ be closed convex subsets of $\mathcal{H}$ such that $\bigcap_{i=1}^m C_i \neq \emptyset$.

We want to

$$\text{Find } \hat{x} \in \bigcap_{i=1}^m C_i.$$

important (if not $\rightarrow$ no CV)

POCS (Projection Onto Convex Sets) algorithm

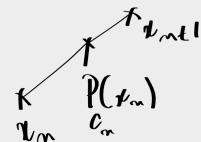
Let $(\lambda_n)_{n \geq 0}$ be a sequence of $[\epsilon_1, 2 - \epsilon_2]$ with $(\epsilon_1, \epsilon_2) \in]0, +\infty[^2$ such that $\epsilon_1 + \epsilon_2 < 2$.

For every $n \in \mathbb{N}$, let $i_n - 1$ denote the remainder after division of n by m .

Set $x_0 \in \mathcal{H}$

For $n = 0, \dots$

$$\lfloor x_{n+1} = x_n + \lambda_n (P_{C_{i_n}}(x_n) - x_n).$$



Convergence of POCS

Theorem

The sequence $(x_n)_{n \in \mathbb{N}}$ generated by the POCS algorithm converges to a point in $\bigcap_{i=1}^m C_i$.

Exercise 1

Let $\mathcal{H}$ be a real Hilbert space.

1. Let $c \in \mathcal{H}$ and $\rho \in]0, +\infty[$. What is the expression of the projection onto a closed ball $\overline{B}(c, \rho)$ with center c and radius ρ ?
2. We consider three closed balls which are assumed to have a common point. Propose an algorithms to compute such a point.

Lagrange duality

Constrained optimization problem

Let $\mathcal{H}$ be a Hilbert space. Let $f: \mathcal{H} \rightarrow]-\infty, +\infty]$.

Let $(m, q) \in \mathbb{N}^2$. For every $i \in \{1, \dots, m\}$, let $g_i: \mathcal{H} \rightarrow \mathbb{R}$ and for every $j \in \{1, \dots, q\}$, let $h_j: \mathcal{H} \rightarrow \mathbb{R}$.

Let

$$C = \{x \in \mathcal{H} \mid (\forall i \in \{1, \dots, m\}) \ g_i(x) = 0 \\ (\forall j \in \{1, \dots, q\}) \ h_j(x) \leq 0\}.$$

We want to:

$$\text{Find } \hat{x} \in \underset{x \in C}{\text{Argmin}} \ f(x).$$

Remark: A vector $x \in \mathcal{H}$ is said to be **feasible** if $x \in \text{dom } f \cap C$.

Definitions

The **Lagrange function** (or Lagrangian) associated with the previous problem is defined as

$$(\forall x \in \mathcal{H})(\forall \mu = (\mu_i)_{1 \leq i \leq m} \in \mathbb{R}^m)(\forall \lambda = (\lambda_j)_{1 \leq j \leq q} \in [0, +\infty[^q)$$

non negative

$$\mathcal{L}(x, \mu, \lambda) = f(x) + \sum_{i=1}^m \mu_i g_i(x) + \sum_{j=1}^q \lambda_j h_j(x).$$

The vectors μ and λ are called **Lagrange multipliers**.

Remark: $\text{dom } \mathcal{L} = \text{dom } f \times \mathbb{R}^m \times [0, +\infty[^q$.

Saddle points

$(\hat{x}, \hat{\mu}, \hat{\lambda}) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$ is a **saddle point** of $\mathcal{L}$ if

$$(\forall (x, \mu, \lambda) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q) \quad \mathcal{L}(\hat{x}, \mu, \lambda) \leq \mathcal{L}(\hat{x}, \hat{\mu}, \hat{\lambda}) \leq \mathcal{L}(x, \hat{\mu}, \hat{\lambda}).$$

Saddle points

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Theorem

Let $\underline{\mathcal{L}}$ and $\overline{\mathcal{L}}$ be defined as

$$\begin{aligned} (\forall (\mu, \lambda) \in \mathbb{R}^m \times [0, +\infty[^q) \quad \underline{\mathcal{L}}(\mu, \lambda) &= \inf_{x \in \mathcal{H}} \mathcal{L}(x, \mu, \lambda) \\ (\forall x \in \mathcal{H}) \quad \overline{\mathcal{L}}(x) &= \sup_{\mu \in \mathbb{R}^m, \lambda \in [0, +\infty[^q} \mathcal{L}(x, \mu, \lambda). \end{aligned}$$

$(\hat{x}, \hat{\mu}, \hat{\lambda}) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$ is a saddle point of $\mathcal{L}$ if and only if

$$\begin{aligned} (\forall x \in \mathcal{H}) \quad \overline{\mathcal{L}}(\hat{x}) &\leq \overline{\mathcal{L}}(x) \\ (\forall (\mu, \lambda) \in \mathbb{R}^m \times [0, +\infty[^q) \quad \underline{\mathcal{L}}(\mu, \lambda) &\leq \underline{\mathcal{L}}(\hat{\mu}, \hat{\lambda}) \\ \underline{\mathcal{L}}(\hat{\mu}, \hat{\lambda}) &= \overline{\mathcal{L}}(\hat{x}). \end{aligned}$$

Saddle points

Theorem

Let $\underline{\mathcal{L}}$ and $\overline{\mathcal{L}}$ be defined as

$$(\forall (\mu, \lambda) \in \mathbb{R}^m \times [0, +\infty[^q) \quad \underline{\mathcal{L}}(\mu, \lambda) = \inf_{x \in \mathcal{H}} \mathcal{L}(x, \mu, \lambda)$$

$$(\forall x \in \mathcal{H}) \quad \overline{\mathcal{L}}(x) = \sup_{\mu \in \mathbb{R}^m, \lambda \in [0, +\infty[^q} \mathcal{L}(x, \mu, \lambda).$$

$(\hat{x}, \hat{\mu}, \hat{\lambda}) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$ is a saddle point of $\mathcal{L}$ if and only if

$$(\forall x \in \mathcal{H}) \quad \overline{\mathcal{L}}(\hat{x}) \leq \overline{\mathcal{L}}(x)$$

$$(\forall (\mu, \lambda) \in \mathbb{R}^m \times [0, +\infty[^q) \quad \underline{\mathcal{L}}(\mu, \lambda) \leq \underline{\mathcal{L}}(\hat{\mu}, \hat{\lambda})$$

$$\underline{\mathcal{L}}(\hat{\mu}, \hat{\lambda}) = \overline{\mathcal{L}}(\hat{x}).$$

Remark: $\overline{\mathcal{L}}$ is called the **primal Lagrange function** and $\underline{\mathcal{L}}$ the **dual Lagrange function**.

Sufficient condition for a constrained minimum

Assume that there exists a feasible point.

If $(\hat{x}, \hat{\mu}, \hat{\lambda}) \in \mathcal{H} \times \mathbb{R}^m \times [0, +\infty[^q$ is a saddle point of $\mathcal{L}$,
then $\hat{x}$ is minimizer of f over C .

In addition, $f(\hat{x}) = \underline{\mathcal{L}}(\hat{\mu}, \hat{\lambda})$ and the **complementary slackness** condition holds:

$$(\forall j \in \{1, \dots, q\}) \quad \hat{\lambda}_j h_j(\hat{x}) = 0.$$

Convex case

Assume that f is a convex function, $(g_i)_{1 \leq i \leq m}$ are affine functions and $(h_j)_{1 \leq j \leq q}$ are convex functions. Assume that the Slater condition holds, i.e. there exists $\bar{x} \in \text{int}(\text{dom } f)$ such that

$$(\forall i \in \{1, \dots, m\}) \quad g_i(\bar{x}) = 0$$

$$(\forall j \in \{1, \dots, q\}) \quad h_j(\bar{x}) < 0.$$

$\hat{x}$ is a minimizer of f over C if and only if there exists $\hat{\mu} \in \mathbb{R}^m$ and $\hat{\lambda} \in [0, +\infty[^q$ such that $(\hat{x}, \hat{\mu}, \hat{\lambda})$ is a saddle point of the Lagrangian.

Important

Convex case

Assume that f is a convex function, $(g_i)_{1 \leq i \leq m}$ are affine functions and $(h_j)_{1 \leq j \leq q}$ are convex functions. Assume that the Slater condition holds, i.e. there exists $\bar{x} \in \text{int}(\text{dom } f)$ such that

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Remark: Under the assumptions of the above theorem, if $\hat{x}$ is a minimizer of f over C then $\mathcal{L}(\cdot, \hat{\mu}, \hat{\lambda})$ is a convex function which is minimum at $\hat{x}$. This optimality condition is often used to calculate $\hat{x}$, in conjunction with the complementary slackness condition.

Convex case

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$\hat{x}$ is a minimizer of f over C if and only if there exists $\hat{\mu} \in \mathbb{R}^m$ and $\hat{\lambda} \in [0, +\infty[^q$ such that $(\hat{x}, \hat{\mu}, \hat{\lambda})$ is a saddle point of the Lagrangian.

Example: Let $c \in \mathcal{H}$ and $\rho \in]0, +\infty[$. Let $C = \overline{B}(c, \rho)$. Let $y \in \mathcal{H} \setminus C$.

We want to find $P_C(y)$. We have thus $m = 0$, $q = 1$, and

$$x \in \overline{B}(c, \rho) \\ \Leftrightarrow \|x - c\| \leq \rho$$

$$(\forall x \in \mathcal{H}) \quad f(x) = \|x - y\|^2 \\ h_1(x) = \|x - c\|^2 - \rho^2.$$

on prend x en dehors
de C car si $y \in C$
alors $P_C(y) = y$

We know that there exists a unique solution $\hat{x}$.

Convex case

Assume that f is a convex function, $(g_i)_{1 \leq i \leq m}$ are affine functions and $(h_j)_{1 \leq j \leq q}$ are convex functions. Assume that the Slater condition holds, i.e. there exists $\bar{x} \in \text{int}(\text{dom } f)$ such that

$$\begin{aligned} (\forall i \in \{1, \dots, m\}) \quad & g_i(\bar{x}) = 0 \\ (\forall j \in \{1, \dots, q\}) \quad & h_j(\bar{x}) < 0. \end{aligned}$$

$\hat{x}$ is a minimizer of f over C if and only if there exists $\hat{\mu} \in \mathbb{R}^m$ and $\hat{\lambda} \in [0, +\infty[^q$ such that $(\hat{x}, \hat{\mu}, \hat{\lambda})$ is a saddle point of the Lagrangian.

Example: Let $c \in \mathcal{H}$ and $\rho \in]0, +\infty[$. Let $C = \overline{B}(c, \rho)$. Let $y \in \mathcal{H} \setminus C$.

We want to find $P_C(y)$. We have thus $m = 0$, $q = 1$, and

$$\begin{aligned} (\forall x \in \mathcal{H}) \quad & f(x) = \|x - y\|^2 \\ & h_1(x) = \|x - c\|^2 - \rho^2. \end{aligned}$$

f and h_1 convex since $\|\cdot\|^2$
(assumptions fct ok)
(Slater condition?)
(yes: $\bar{x} = c$ (center of the ball C))

Let $\bar{x} = c$. Then $\bar{x} \in \text{int}(\text{dom } f) = \mathcal{H}$ and $h_1(\bar{x}) = -\rho^2 < 0$. Therefore

Slater condition holds.

Convex case

Example: Let $c \in \mathcal{H}$ and $\rho \in]0, +\infty[$. Let $C = \overline{B}(c, \rho)$. Let $y \in \mathcal{H} \setminus C$. We want to find $P_C(y)$. We have thus $m = 0$, $q = 1$, and

$$\begin{aligned} (\forall x \in \mathcal{H}) \quad f(x) &= \|x - y\|^2 \\ h_1(x) &= \|x - c\|^2 - \rho^2. \end{aligned}$$

Let $\bar{x} = c$. Then $\bar{x} \in \text{int}(\text{dom } f) = \mathcal{H}$ and $h_1(\bar{x}) = -\rho^2 < 0$. Therefore Slater condition holds.

There thus exists $\hat{\lambda} \in [0, +\infty[$ such that $(\hat{x}, \hat{\lambda})$ is a saddle point of

$$(\forall x \in \mathcal{H})(\forall \lambda \in [0, +\infty[) \quad \mathcal{L}(x, \lambda) = f(x) + \lambda h_1(x).$$

Since $\hat{x}$ minimizes $\mathcal{L}(\cdot, \hat{\lambda})$, the first-order optimality condition yields

$$\nabla_x \mathcal{L}(\hat{x}, \hat{\lambda}) = 2(\hat{x} - y) + 2\hat{\lambda}(\hat{x} - c) = 0 \quad \Leftrightarrow \quad \hat{x} = \frac{y + \hat{\lambda}c}{1 + \hat{\lambda}}.$$

In addition, it follows from the complementary slackness condition that $\hat{\lambda}h_1(\hat{x}) = 0$. Either $\hat{\lambda} = 0 \Rightarrow \hat{x} = y$: impossible since $y \notin C$ or $\hat{\lambda} \neq 0 \Rightarrow h_1(\hat{x}) = 0 \Leftrightarrow \hat{\lambda} = \|y - c\|/\rho - 1 \Leftrightarrow \hat{x} = c + \rho \frac{y - c}{\|y - c\|}$.

Convex case

Assume that f is a convex function, $(g_i)_{1 \leq i \leq m}$ are affine functions and $(h_j)_{1 \leq j \leq q}$ are convex functions. Assume that the Slater condition holds, i.e. there exists $\bar{x} \in \text{int}(\text{dom } f)$ such that

$$\begin{aligned} (\forall i \in \{1, \dots, m\}) \quad & g_i(\bar{x}) = 0 \\ (\forall j \in \{1, \dots, q\}) \quad & h_j(\bar{x}) < 0. \end{aligned}$$

$\hat{x}$ is a minimizer of f over C if and only if there exists $\hat{\mu} \in \mathbb{R}^m$ and $\hat{\lambda} \in [0, +\infty[^q$ such that $(\hat{x}, \hat{\mu}, \hat{\lambda})$ is a saddle point of the Lagrangian.

Remark: Alternatively, $(\hat{\mu}, \hat{\lambda}) = \underset{\mu \in \mathbb{R}^m, \lambda \in [0, +\infty[^q}{\operatorname{argmax}} \mathcal{L}(\mu, \lambda)$

and $\hat{x} = \underset{x \in \mathcal{H}}{\operatorname{argmin}} \mathcal{L}(x, \hat{\mu}, \hat{\lambda})$.

Differentiable case

Karush-Kuhn-Tucker (KKT) theorem

Assume that f , $(g_i)_{1 \leq i \leq m}$, and $(h_j)_{1 \leq j \leq q}$ are continuously differentiable on $\mathcal{H} = \mathbb{R}^N$.

Assume that $\hat{x}$ is a local minimizer of f over C satisfying the following Mangasarian-Fromovitz **constraint qualification conditions** :

- (i) $\{\nabla g_i(\hat{x}) \mid i \in \{1, \dots, m\}\}$ is a family of linearly independent vectors;
- (ii) there exists $z \in \mathbb{R}^N$ such that

$$\begin{aligned} (\forall i \in \{1, \dots, m\}) \quad & \langle \nabla g_i(\hat{x}) \mid z \rangle = 0 \\ (\forall j \in J(\hat{x})) \quad & \langle \nabla h_j(\hat{x}) \mid z \rangle < 0 \end{aligned}$$

where $J(\hat{x}) = \{j \in \{1, \dots, q\} \mid h_j(\hat{x}) = 0\}$ is the set of **active inequality constraints** at $\hat{x}$.

Then, there exists $\hat{\mu} \in \mathbb{R}^N$ and $\hat{\lambda} \in [0, +\infty[^q$ such that $\hat{x}$ is a critical point of $\mathcal{L}(\cdot, \hat{\mu}, \hat{\lambda})$ and the complementary slackness condition holds.

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- (ii) there exists $z \in \mathbb{R}^N$ such that

$$\begin{aligned} (\forall i \in \{1, \dots, m\}) \quad & \langle \nabla g_i(\hat{x}) \mid z \rangle = 0 \\ (\forall j \in J(\hat{x})) \quad & \langle \nabla h_j(\hat{x}) \mid z \rangle < 0. \end{aligned}$$

Then, there exists $\hat{\mu} \in \mathbb{R}^m$ and $\hat{\lambda} \in [0, +\infty[^q$ such that $\hat{x}$ is a critical point of $\mathcal{L}(\cdot, \hat{\mu}, \hat{\lambda})$ and the complementary slackness condition holds.

Remark: A sufficient condition for **Mangasarian-Fromovitz conditions** to be satisfied is that $\{\nabla g_i(\hat{x}) \mid i \in \{1, \dots, m\}\} \cup \{\nabla h_j(\hat{x}) \mid j \in J(\hat{x})\}$ is a family of linearly independent vectors.

Differentiable case

Example: Solve the following problem:

$$\underset{x=(x^{(i)})_{1 \leq i \leq N} \in S}{\text{maximize}} \quad (x^{(N)})^3 - \frac{1}{2}(x^{(N)})^2$$

where S is the unit sphere, centered at 0, of $\mathbb{R}^N$ with $N \geq 2$.

Existence of a solution guaranteed by Weierstrass theorem.

- ▶ $(\forall x = (x^{(i)})_{1 \leq i \leq N} \in \mathbb{R}^N) \quad f(x) = \frac{1}{2}(x^{(N)})^2 - (x^{(N)})^3$
- ▶ $m = 1, \quad q = 0$
- ▶ $(\forall x \in \mathbb{R}^N) \quad g_1(x) = \|x\|^2 - 1.$

Let $\hat{x}$ be a solution. $\nabla g_1(\hat{x}) = 2\hat{x}$.

$\rightsquigarrow$ Mangasarian-Fromovitz conditions are satisfied if $\hat{x} \neq 0$.

Lagrange function:

$$(\forall x \in \mathbb{R}^N)(\forall \mu \in \mathbb{R}) \quad \mathcal{L}(x, \mu) = f(x) + \mu(\|x\|^2 - 1).$$

Differentiable case

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Lagrange function:

$$(\forall x \in \mathbb{R}^N)(\forall \mu \in \mathbb{R}) \quad \mathcal{L}(x, \mu) = \underbrace{f(x)}_{x^{(N)}} + \mu \underbrace{\left(\|x\|^2 - 1 \right)}_{\sum_{i=1}^N (x^{(i)})^2}.$$

According to KKT theorem, if $\hat{x} = (\hat{x}^{(i)})_{1 \leq i \leq N} \neq 0$, then there exists $\hat{\mu} \in \mathbb{R}$ such that $\nabla_x \mathcal{L}(\hat{x}, \hat{\mu}) = 0$, i.e.

$$\begin{aligned} (\forall i \in \{1, \dots, N-1\}) \quad \hat{\mu} \hat{x}^{(i)} &= 0 \\ \hat{x}^{(N)} - 3(\hat{x}^{(N)})^2 + 2\hat{\mu} \hat{x}^{(N)} &= 0 \end{aligned}$$

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According to KKT theorem, if $\hat{x} = (\hat{x}^{(i)})_{1 \leq i \leq N} \neq 0$, then there exists $\hat{\mu} \in \mathbb{R}$ such that $\nabla_x \mathcal{L}(\hat{x}, \hat{\mu}) = 0$, i.e.

$$\begin{aligned} (\forall i \in \{1, \dots, N-1\}) \quad \hat{\mu} \hat{x}^{(i)} &= 0 \\ \hat{x}^{(N)} - 3(\hat{x}^{(N)})^2 + 2\hat{\mu} \hat{x}^{(N)} &= 0 \end{aligned}$$

Either $\hat{\mu} \neq 0$ and $(\forall i \in \{1, \dots, N-1\}) \hat{x}^{(i)} = 0 \Rightarrow \hat{x}^{(N)} = \pm 1$, $\sum_{i=1}^N (x^{(i)})^2 = 1$
or $\hat{\mu} = 0$ and $\hat{x}^{(N)} - 3(\hat{x}^{(N)})^2 = 0$, that is $\hat{x}^{(N)} = 0$ or $\hat{x}^{(N)} = 1/3$.

By looking at the minimum cost solution, we deduce that $\hat{x} = (0, \dots, 0, 1)$.

Exercise 2

Let f be defined as

$$(\forall x = (x^{(i)})_{1 \leq i \leq N} \in \mathbb{R}^N) \quad f(x) = \sum_{i=1}^N \exp(x^{(i)})$$

with $N > 1$. We want to find a minimizer of f on $\mathbb{R}^N$ subject to the constraints

$$\sum_{i=1}^N x^{(i)} = 1$$

$$(\forall i \in \{1, \dots, N\}) \quad x^{(i)} \geq 0.$$

1. What can be said about the existence/uniqueness of a solution to this problem ?
2. Apply the Lagrange multiplier method.

A few algorithms

Problem

Let $\mathcal{H}$ be a Hilbert space.

Let $f: \mathcal{H} \rightarrow \mathbb{R}$ be differentiable.

Let C be a nonempty closed convex subset of $\mathcal{H}$.

We want to:

$$\text{Find } \hat{x} \in \underset{x \in C}{\text{Argmin}} f(x).$$

Problem

Let $\mathcal{H}$ be a Hilbert space.

Let $f: \mathcal{H} \rightarrow \mathbb{R}$ be differentiable.

Let C be a nonempty closed convex subset of $\mathcal{H}$.

We want to:

$$\text{Find } \hat{x} \in \underset{x \in C}{\text{Argmin}} f(x).$$

Objective: Build a sequence $(x_n)_{n \in \mathbb{N}}$ converging to a minimizer.

Principle of first-order methods

- ▶ If f is differentiable, then, at iteration n , we have

$$(\forall x \in \mathcal{H}) \quad f(x) = f(x_n) + \langle \nabla f(x_n) \mid x - x_n \rangle + o(\|x - x_n\|).$$

So if $\|x_{n+1} - x_n\|$ is small enough and x_{n+1} is chosen such that

$$\langle \nabla f(x_n) \mid x_{n+1} - x_n \rangle < 0$$

then $f(x_{n+1}) < f(x_n)$.

- ▶ In particular, the **steepest descent direction** is given by

$$x_{n+1} - x_n = -\gamma_n \nabla f(x_n), \quad \gamma_n \in]0, +\infty[.$$

- ▶ To secure that the solution belongs to C we can add a projection step.
- ▶ A relaxation parameter λ_n can also be added.

Principle of first-order methods *most famous algo*

The **projected gradient algorithm** has the following form:

$$x_{n+1} = x_n + \lambda_n (P_C(x_n - \gamma_n \nabla f(x_n)) - x_n)$$

where $\gamma_n \in]0, +\infty[$ and $\lambda_n \in]0, 1]$.

Principle of first-order methods

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Remark: x is a fixed point of the projected gradient iteration if and only if $x \in C$ and

$$(\forall y \in C) \quad \langle \nabla f(x) \mid y - x \rangle \geq 0.$$

Principle of first-order methods

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Remark: x is a fixed point of the projected gradient iteration if and only if $x \in C$ and

$$(\forall y \in C) \quad \langle \nabla f(x) \mid y - x \rangle \geq 0.$$

Proof: If x is a fixed point, then

$$\begin{aligned} x &= x + \lambda_n (P_C(x - \gamma_n \nabla f(x)) - x) \\ \Leftrightarrow x &= P_C(x - \gamma_n \nabla f(x)). \end{aligned}$$

According to the characterization of the projection, for every $y \in C$,

$$\begin{aligned} \langle x - \gamma_n \nabla f(x) - x \mid y - x \rangle &\leq 0 \\ \Leftrightarrow \langle \nabla f(x) \mid y - x \rangle &\geq 0. \end{aligned}$$

Principle of first-order methods

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where $\gamma_n \in]0, +\infty[$ and $\lambda_n \in]0, 1]$.

Remark:

- ▶ x is a fixed point of the projected gradient iteration if and only if $x \in C$ and

$$(\forall y \in C) \quad \langle \nabla f(x) \mid y - x \rangle \geq 0.$$

- ▶ When f is convex, x is a fixed point of the projected gradient iteration if and only if x is a global minimizer of f over C .

Principle of first-order methods

The **projected gradient algorithm** has the following form:

$$x_{n+1} = x_n + \lambda_n (P_C(x_n - \gamma_n \nabla f(x_n)) - x_n)$$

where $\gamma_n \in]0, +\infty[$ and $\lambda_n \in]0, 1]$.

Remark: If $C = \mathcal{H}$ and $\lambda_n = 1$, we recover the standard gradient descent iteration:

$$x_{n+1} = x_n - \gamma_n \nabla f(x_n).$$

Convergence

Convergence theorem

Assume that f is convex and has a Lipschitzian gradient with constant $\beta \in]0, +\infty[$, i.e.

$$(\forall (x, y) \in \mathcal{H}^2) \quad \|\nabla f(x) - \nabla f(y)\| \leq \beta \|x - y\|.$$

Assume that $\text{Argmin}_{x \in C} f(x) \neq \emptyset$.

Assume that $\inf_{n \in \mathbb{N}} \gamma_n > 0$, $\sup_{n \in \mathbb{N}} \gamma_n < 2/\beta$, $\inf_{n \in \mathbb{N}} \lambda_n > 0$, and $\sup_{n \in \mathbb{N}} \lambda_n \leq 1$.

Then the sequence $(x_n)_{n \in \mathbb{N}}$ generated by the projected gradient algorithm converges to a minimizer of f over C .

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Then the sequence $(x_n)_{n \in \mathbb{N}}$ generated by the projected gradient algorithm converges to a minimizer of f over C .

In addition, if f is strongly convex, the convergence is linear, i.e. there exists $\chi \in [0, 1[$ such that

$$(\forall n \in \mathbb{N}) \quad \|x_n - \hat{x}\| \leq \chi^n \|x_0 - \hat{x}\|$$

where $\hat{x}$ is the unique minimizer of f over C .

Convergence

Convergence theorem

Assume that f is convex and has a Lipschitzian gradient with constant $\beta \in]0, +\infty[$.

Assume that $\text{Argmin}_{x \in C} f(x) \neq \emptyset$.

Assume that $\inf_{n \in \mathbb{N}} \gamma_n > 0$, $\sup_{n \in \mathbb{N}} \gamma_n < 2/\beta$, $\inf_{n \in \mathbb{N}} \lambda_n > 0$, and $\sup_{n \in \mathbb{N}} \lambda_n \leq 1$.

Then the sequence $(x_n)_{n \in \mathbb{N}}$ generated by the projected gradient algorithm converges to a minimizer of f over C .

Remark: If f is non convex with a β -Lipschitzian gradient, it can only be proved that $(f(x_n))_{n \in \mathbb{N}}$ is a converging sequence provided that $\gamma_n \leq 2/\beta$.

Example: Uzawa algorithm

Problem

Let $\mathcal{L}: \mathcal{H} \times [0, +\infty[^q \rightarrow \mathbb{R}$ be differentiable with respect to its second argument. We want to find a saddle point of $\mathcal{L}$

Solution

Set $\lambda_0 \in [0, +\infty[^q$

For $n = 1, \dots$

$$\left\{ \begin{array}{l} \text{Set } \gamma_n \in]0, +\infty[, \rho_n \in]0, 1] \\ x_n \in \text{Argmin} \mathcal{L}(\cdot, \lambda_n) \\ \lambda_{n+1} = \lambda_n + \rho_n (P_{[0, +\infty[^q}(\lambda_n + \gamma_n \nabla_{\lambda} \mathcal{L}(x_n, \lambda_n)) - \lambda_n). \end{array} \right.$$

Frank-Wolfe algorithm

- ▶ Motivation
Complexity of projected gradient when P_C is complicated (e.g. a polyhedron).
- ▶ Search direction

$$x_{n+1} - x_n = \gamma_n(z_n - x_n)$$

where

$$z_n \in \underset{z \in C}{\operatorname{Argmin}} \langle \nabla f(x_n) \mid z - x_n \rangle \stackrel{\text{H. Weinstan}}{=} \underset{z \in C}{\operatorname{Argmin}} \langle \nabla f(x_n) \mid z \rangle$$

and $\gamma_n \in]0, 1]$.

- ▶ Feasibility

If C is a nonempty compact convex set and $x_0 \in C$, then

$x_{n+1} = (1 - \gamma_n)x_n + \gamma_n z_n$ is a well-defined element of C .

Convergence rate

Theorem

Let $f: \mathcal{H} \rightarrow \mathbb{R}$ be a differentiable convex function with a Lipschitzian gradient. Let C be a nonempty compact convex set.

Assume that a minimizer $\hat{x}$ of f over C exists.

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence generated by the Frank-Wolfe algorithm with $x_0 \in C$ and

$$(\forall n \in \mathbb{N}) \quad \gamma_n = \frac{2}{n+2}.$$

Then, there exists $\eta > 0$ such that

$$(\forall n \in \mathbb{N}) \quad f(x_n) - f(\hat{x}) \leq \frac{\eta}{n+2}.$$

Principle of second-order methods

- ▶ If f is twice differentiable, then, at iteration n , we have

$$\begin{aligned} (\forall x \in \mathcal{H}) \quad f(x) = & f(x_n) + \langle \nabla f(x_n) \mid x - x_n \rangle \\ & + \frac{1}{2} \langle (x - x_n) \mid \nabla^2 f(x_n)(x - x_n) \rangle + o(\|x - x_n\|^2). \end{aligned}$$

- ▶ If $\nabla^2 f(x_n)$ is positive definite, the minimizer x_{n+1} of the quadratic term is given by **Newton's iteration**

$$\begin{aligned} \nabla f(x_n) + \nabla^2 f(x_n)(x_{n+1} - x_n) &= 0 \\ \Leftrightarrow x_{n+1} &= x_n - (\nabla^2 f(x_n))^{-1} \nabla f(x_n). \end{aligned}$$

Convergence

Convergence theorem

Let $f \in \mathcal{H} \rightarrow]-\infty, +\infty]$ be three times continuously differentiable in a neighborhood of a local minimizer $\hat{x}$ and assume that $\nabla^2 f(\hat{x})$ is positive definite.

Then, there exists $\epsilon \in]0, +\infty[$ such that, if $\|x_0 - \hat{x}\| \leq \epsilon$, then $(x_n)_{n \in \mathbb{N}}$ converges to $\hat{x}$.

In addition, the convergence is quadratic, i.e. there exists $\kappa \in]0, +\infty[$ such that

$$(\forall n \in \mathbb{N}) \quad \|x_{n+1} - \hat{x}\| \leq \kappa \|x_n - \hat{x}\|^2.$$

Numerical behaviour

- ▶ Although the convergence of Newton's algorithm is faster than the gradient descent in terms of iteration number, the computational cost of each iteration is higher.
- ▶ To improve the convergence guarantees of Newton's algorithm, we may practically modify it as follows:

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n (\nabla^2 f(x_n) + \lambda_n \text{Id})^{-1} \nabla f(x_n),$$

with $(\gamma_n, \lambda_n) \in]0, +\infty[^2$.

- ▶ Quasi-Newton algorithms read

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - H_n^{-1} \nabla f(x_n),$$

where H_n is a definite positive matrix providing some approximation to the Hessian.

Subspace algorithms

► New update rule

$$(\forall n \in \mathbb{N}) \quad x_{n+1} - x_n \in \text{Span}\{\nabla f(x_n), d_n\}$$

where $d_n \in \mathcal{H}$ search direction, e.g.

$$d_n = x_n - x_{n-1} \quad (\text{memory gradient}).$$

Subspace algorithms

- ▶ New update rule

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n^{(1)} \nabla f(x_n) + \underbrace{(\gamma_n^{(2)} d_n)}_{\text{added part for the new direction}}$$

where d_n search direction and $(\gamma_n^{(1)}, \gamma_n^{(2)}) \in \mathbb{R}^2$ step-sizes.

Subspace algorithms

► New update rule

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n^{(1)} \nabla f(x_n) + \gamma_n^{(2)} d_n$$

where d_n search direction and $(\gamma_n^{(1)}, \gamma_n^{(2)}) \in \mathbb{R}^2$ step-sizes.

► Search for the step-sizes

$$(\gamma_n^{(1)}, \gamma_n^{(2)}) \in \underset{(\gamma^{(1)}, \gamma^{(2)}) \in \mathbb{R}^2}{\text{Argmin}} \quad f(x_n - \gamma^{(1)} \nabla f(x_n) + \gamma^{(2)} d_n)$$

↪ often too time consuming

↪ set $\gamma_n^{(2)} = \gamma_n^{(1)} \alpha_n$ where α_n is given by some heuristic rule and optimize $\gamma_n^{(1)}$ (conjugate gradient method)

Subspace algorithms

- ▶ New update rule

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n^{(1)} \nabla f(x_n) + \gamma_n^{(2)} d_n$$

where d_n search direction and $(\gamma_n^{(1)}, \gamma_n^{(2)}) \in \mathbb{R}^2$ step-sizes.

- ▶ Search for the step-sizes

$$(\gamma_n^{(1)}, \gamma_n^{(2)}) \in \underset{(\gamma^{(1)}, \gamma^{(2)}) \in \mathbb{R}^2}{\text{Argmin}} \quad f(x_n - \gamma^{(1)} \nabla f(x_n) + \gamma^{(2)} d_n)$$

$\rightsquigarrow$ alternative strategy:

Use a surrogate function, e.g. a quadratic function, for which optimum values of $(\gamma_n^{(1)}, \gamma_n^{(2)})$ have a closed form expression.

Subspace algorithms

► New update rule

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n - \gamma_n^{(1)} \nabla f(x_n) + \gamma_n^{(2)} d_n$$

where d_n search direction and $(\gamma_n^{(1)}, \gamma_n^{(2)}) \in \mathbb{R}^2$ step-sizes.

► Search for the step-sizes

$$(\gamma_n^{(1)}, \gamma_n^{(2)}) \in \underset{(\gamma^{(1)}, \gamma^{(2)}) \in \mathbb{R}^2}{\text{Argmin}} \quad f(x_n - \gamma^{(1)} \nabla f(x_n) + \gamma^{(2)} d_n)$$

↪ alternative strategy:

Use a surrogate function, e.g. a quadratic function, for which optimum values of $(\gamma_n^{(1)}, \gamma_n^{(2)})$ have a closed form expression.

Remark: Higher dimensional subspaces can be used (e.g., L-BFGS method)

Exercise 3

Let $\mathcal{H}$ and $\mathcal{G}$ be real Hilbert spaces and let $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$. Let $y \in \mathcal{G}$ and let $\alpha \in]0, +\infty[$.

We want to minimize the function defined as

$$(\forall x \in \mathcal{H}) \quad f(x) = \frac{1}{2} \|Lx - y\|^2 + \frac{\alpha}{2} \|x\|^2.$$

1. Give the form of the gradient descent algorithm allowing us to solve this problem.
2. How does Newton's method read for this function ?
3. Consider the case when $\mathcal{H} = \mathbb{R}^N$. Study the convergence of the gradient descent algorithm by performing the eigendecomposition of L^*L .

Exercise 3:

1) f is strongly convex and coercive.

So f has a unique minimizer.

$$\nabla f(x) = \underset{\substack{\uparrow \\ \text{adjoint of } L}}{L^*}(Lx - y) + \alpha x$$

$$\nabla \varphi \circ L(x) = L^* \nabla \varphi(Lx)$$

$$x_{n+1} = x_n - \delta_n \nabla f(x_n) = x_n - \delta_n [L^*(Lx_n - y) + \alpha x_n] \quad \delta_n > 0$$

$$\forall (x, z) \in \mathbb{H}^2, \quad \|\nabla f(x) - \nabla f(z)\| = \|(L^*L + \alpha I)(x - z)\|$$
$$\leq \underbrace{\|L^*L + \alpha I\|}_{\beta \text{ (Lipschitzienne)}} \|x - z\|$$

($L^*L \rightarrow$ semi-definite positive)

$\delta_n < \frac{2}{\beta} \rightarrow \text{CV of the alg.}$

$$\langle x, L^* L x \rangle \stackrel{\downarrow}{=} \langle Lx, Lx \rangle = \|Lx\|^2 > 0$$

$$\text{So } \beta = \|L^* L\| + \alpha = \|L\|^2 + \alpha$$

(démonstration avec la norme spectrale)

$$\nabla^2 f(x) = \underbrace{L^* L}_{\geq 0} + \alpha I > 0$$

$$x_{n+1} = x_n - [\nabla^2 f(x_n)]^{-1} \nabla f(x_n)$$

$$= x_n - (L^* L + \alpha I)^{-1} [(L^* L + \alpha I) x_n - L^* y]$$

$$= (L^* L + \alpha I)^{-1} L^* y$$

Conseils:

- Prendre un autre cours d'optimization 
(sur le catalogue)
 - ↳ 1 Basique (niveau ok)
 - ↳ l'autre avancé (par Emiliè....)
niveau plus compliqué